# GAMAL FARAG

## DIGITAL ENGINEERING & OPERATIONS | PROJECT CONTROLS

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Project portfolio: https://gamaleldinyousryibrahimfarag.github.io/gamal-portfolio/digital.html

## PROFESSIONAL PROFILE

Civil engineer and engineering-management professional with six years of experience across owner-representative, consultancy and contractor environments. Combines construction project controls with hands-on development of digital applications and AI-enabled workflows for procurement, document control, operational reporting and service discovery. Connects business needs to practical system logic: mapping processes, building tools, validating outputs, troubleshooting issues and supporting day-to-day use across teams.

## WHAT I BRING

**BUILD & AUTOMATE** Develop practical applications and AI-assisted workflows that convert repetitive business tasks into structured digital processes.

**OPERATE & IMPROVE** Validate outputs, resolve data and workflow issues, support users and refine tools for reliable day-to-day operation.

**COORDINATE & DELIVER** Connect site engineering, finance, procurement, document control and management around clear handoffs, responsibilities and records.

## SELECTED SYSTEMS AND PRACTICAL DIGITAL WORK

**Primavera P6 progress-update automation** Developed and tested spreadsheet/script-assisted P6 update workflows involving activity data mapping, progress-percentage normalization, date checks, import preparation and error logging.

**Procurement Management Application** Developed a cross-functional application connecting site engineers, finance/accounting and the procurement lead through a security-focused workflow.

**AI-powered Document Control Application** Developed an AI vision-enabled intake workflow that interprets incoming emails, identifies concerned engineers, routes correspondence, files records and creates a log.

**Management performance systems** Built structured P6/Excel/Power BI reporting around planned vs actual progress, resource and cost loading, S-curves, critical-path exposure and recovery actions.

**Automotive Services Application** Developed service-discovery features covering automotive service categories, vehicle price information and nearby service providers.

**Predictive AI & historical-data modelling** Trained and experimented with AI models using historical datasets to identify patterns and explore forecasts of future outcomes.

## PROFESSIONAL EXPERIENCE

### AL FAISAL HOLDING | Planning Engineer, Owner Representative Office | Mar 2024 – Present

Doha, Qatar | Private Royal Developments / Four Palaces Programme (QAR 200M+)

- Developed and coordinated resource-loaded Primavera P6 baselines of 4,000+ activities across four concurrent projects, integrating civil, MEP, fit-out, external works and procurement interfaces.
- Turned site updates into management information: milestone forecasts, earned-value indicators, critical-path analysis, delay exposure and clear recovery priorities.
- Prepared and improved project-control and procurement-related workflows, linking engineering decisions, package delivery, approvals and construction sequences.
- Coordinated contractors, subcontractors and owner-side stakeholders; reviewed tender programmes, evaluated slippages and supported delay, TIA and EOT assessments.

## PROFESSIONAL EXPERIENCE | CONTINUED

**MANNAI CORPORATION / GULF LABORATORIES | Planning Engineer (Civil) | Sep 2021 – Mar 2024**

Doha, Qatar | ASHGHAL Deep Injection Wells Programme (QAR 150M)

- Integrated three subcontractor programmes into the P6 baseline, connecting execution logic with resources, procurement and infrastructure delivery constraints.
- Produced weekly/monthly earned-value analysis, S-curves, dashboards and corrective planning scenarios; supported time-impact and EOT records.

**CEG INTERNATIONAL | Planning Engineer (Civil) | Nov 2020 – Aug 2021**

Doha, Qatar | ASHGHAL Inner Doha Sewer Programme (QAR 500M)

- Consolidated five subcontractor schedules into a CPM master programme for tunnelling and civil infrastructure works.
- Monitored deviations, manpower and cost performance; supported risk escalation, delay review and recovery planning.

## REPRESENTATIVE ENGINEERING DELIVERY

**FOUR PALACES PROGRAMME | Qatar** Owner-side controls across four parallel royal developments; integrated construction, fit-out, specialist procurement and decision milestones.

**SOUQ AL NAJADA – GOLD SOUQ | Doha** Resource- and cost-aware construction sequencing, performance tracking and recovery logic for a heritage/retail scope.

**LEKHWIYA MULTI-BUILDING PROJECT | Qatar** Steel erection and building-level programmes integrating engineering, procurement, workforce and site sequence.

**DIRIYAH SQUARE – PACKAGE 3 | Riyadh** Procurement and installation planning for specialist architectural packages across 76 buildings.

**ASPIRE ZONE PHASE 2 – WEST ZONE | Qatar** Design-build programme for roads, buildings, athletic facilities, power, lighting, irrigation and landscaping interfaces.

**OFFICE TOWER 404 | Doha Corniche** Planning an 8,303 $m^2$ curtain-wall replacement around occupied-building access and long-lead procurement.

## ENGINEERING, TECHNICAL AND BUSINESS CAPABILITIES

**Digital build and automation** Python, Excel/VBA, application workflow design, scripting and data processing, AI-assisted development (Cursor, ChatGPT, Gemini), AI vision-enabled document workflows, predictive modelling with historical data, testing and iteration.

**Procurement and operations** Procurement application workflows connecting site, finance and purchasing; package sequencing, engineering/submittal milestones, supplier interfaces and contract controls.

**Reporting and engineering tools** Advanced Excel, Power BI, dashboards, AutoCAD, Revit, Microsoft Office.

## EDUCATION, CREDENTIALS AND LANGUAGES

**M.Sc. Engineering Management** Qatar University, 2024 | GPA 3.92/4.0

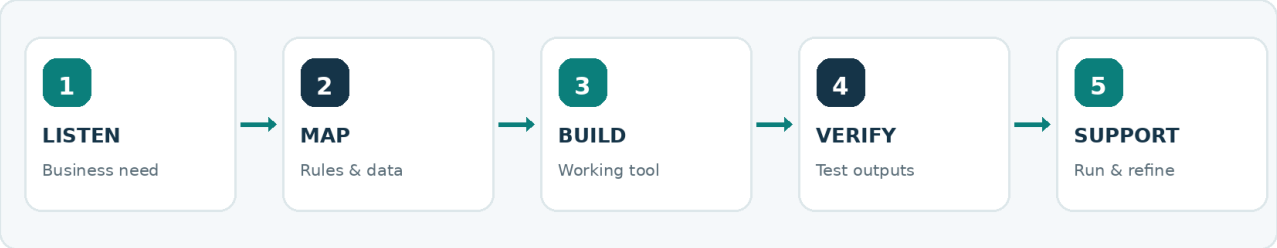
**B.Sc. Civil Engineering** Ain Shams University, 2020

**Professional credentials** UPDA holder; Egyptian Engineers Syndicate; Federation of Arab Engineers. CIOB Professional Review preparation (MCIOB pathway).

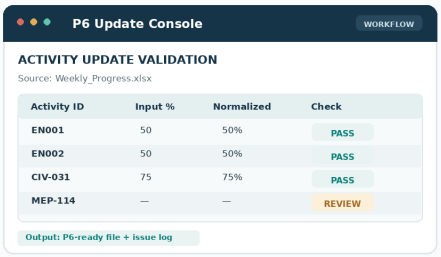
**Languages** Arabic: Native | English: Professional working proficiency (IELTS 7.5)

# TURNING ENGINEERING TASKS INTO AUTOMATION

I translate repetitive engineering and procurement work into structured, testable tools - then help teams use, troubleshoot and improve them.




## SELECTED WORKFLOWS & FEATURES

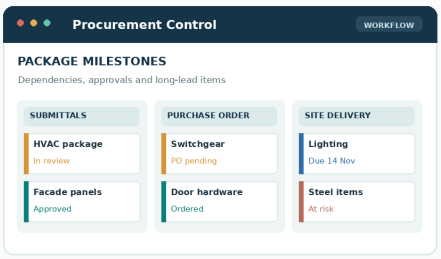



### 01 P6 PROGRESS UPDATE AUTOMATION

**TASK** Repeated weekly activity updates from Excel, with inconsistent dates, IDs and percentages.

**AUTOMATION** Read inputs, check activity IDs, normalize progress and dates, prepare import data and flag errors.

**MY INVOLVEMENT** Built and iterated the update workflow; tested P6 data handling, validation and error logging.

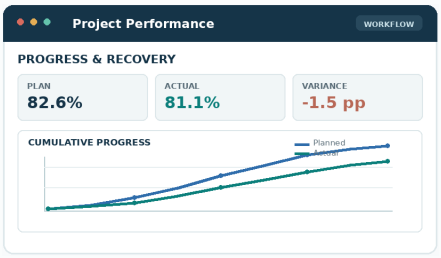



### 02 PROCUREMENT PACKAGE & CONTRACT CONTROL

**TASK** Multiple package approvals, purchasing stages, lead times and site dependencies to coordinate.

**AUTOMATION APPROACH** Structure submittal, PO, payment and delivery gates; monitor package status and exceptions.

**MY INVOLVEMENT** Developed procurement programmes and tracking logic linking supplier commitments to construction readiness.




### 03 PERFORMANCE REPORTING & ACTIONS

**TASK** Turn scattered site, cost and P6 progress information into clear management priorities.

**AUTOMATION APPROACH** Standardize updates, compare plan vs. actual, surface milestone drift and build dashboards.

**MY INVOLVEMENT** Prepared reporting frameworks, reviewed outputs and coordinated recovery priorities with teams.

## HOW I ADD VALUE

**UNDERSTAND**
Translate real site requests into practical requirements

**BUILD & TEST**
Python / Excel / VBA / AI-assisted development, with validation

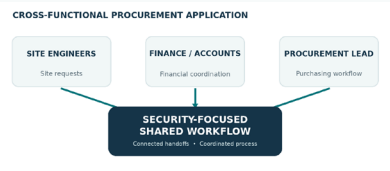
**OPERATE & IMPROVE**
Support users, resolve issues and refine the workflow

# BUILDING APPLICATIONS THAT CONNECT TEAMS & AUTOMATE WORK

From cross-functional procurement workflows to AI-driven correspondence handling and location-based service discovery.

## HOW I TURN A BUSINESS TASK INTO A WORKING DIGITAL PROCESS

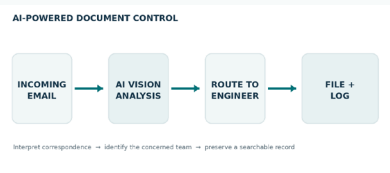
| 01 UNDERSTAND | 02 DESIGN | 03 AUTOMATE | 04 OPERATE |
|---|---|---|---|
| Task & users | Workflow & rules | Application & AI | Test, support, improve |




### 01 PROCUREMENT MANAGEMENT APPLICATION

**CROSS-FUNCTIONAL BUSINESS WORKFLOW**

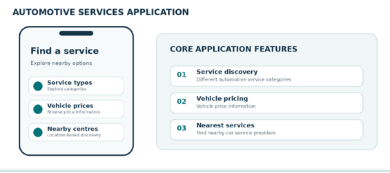
**CONNECTED TEAMS** Site engineers, finance/accounting and the procurement lead work through one coordinated application workflow.
**DESIGN FOCUS** Security-focused process flow connecting stakeholders and their procurement handoffs.
**VALUE** Brings the procurement process and its participating functions into one connected digital environment.




### 02 AI-POWERED DOCUMENT CONTROL APPLICATION

**AI VISION • CORRESPONDENCE ROUTING • RECORDS**

**INTAKE** AI vision model interprets incoming email content and identifies the relevant subject / responsible engineers.
**AUTOMATION** Routes correspondence to concerned engineers, files the related record and creates a log.
**VALUE** Reduces manual document-control handling and supports consistent routing, filing and traceability.




### 03 AUTOMOTIVE SERVICES APPLICATION

**CONSUMER-FACING SERVICE DISCOVERY**

**FEATURES** Automotive service categories and vehicle price information.
**LOCATION** Helps users discover nearby car service providers.
**VALUE** Brings service information, pricing context and location-based discovery into one application.

## 04 PREDICTIVE AI & HISTORICAL-DATA MODELLING




Trained and experimented with AI models on historical datasets to identify past patterns and explore predictions of future outcomes. Applied a data-driven approach to model development and evaluation.